Prayas JEE AIR (2025)

WEEKLY TEST 28

DURATION : 180 Minutes | **DATE : 09/01/2025** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1. (3)
2. (1)
3. (3)
4. (3)
5. (1)
6. (3)
7. (3)
8. (3)
9. (2)
10. (2)
11. (2)
12. (1)
13. (2)
14. (1)
15. (3)
16. (2)
17. (2)
18. (4)
19. (4)
20. (2)
21. (105)
22. (4)
23. (5)
24. (5)
25. (2)

CHEMISTRY

26. (1)
27. (4)
28. (3)
29. (3)
30. (3)
31. (2)
32. (2)
33. (1)
34. (1)
35. (4)
36. (4)
37. (2)
38. (1)
39. (2)
40. (4)
41. (2)
42. (1)
43. (2)
44. (3)
45. (1)
46. (7)
47. (4)
48. (9)
49. (5)
50. (270)

MATHEMATICS

51. (1)
52. (2)
53. (1)
54. (2)
55. (2)
56. (4)
57. (1)
58. (4)
59. (4)
60. (2)
61. (2)
62. (1)
63. (4)
64. (3)
65. (1)
66. (4)
67. (1)
68. (3)
69. (2)
70. (4)
71. (398)
72. (18)
73. (4)
74. (5)
75. (1)

SECTION-I (PHYSICS)

1. **(3)**

The induce emf at t = 1s i.e. at x = 0m is,

$E = B\,l\,v = B(2R)v$

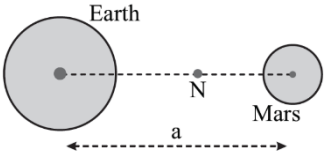
$\Rightarrow E = 1 \times 1 \times 2 \times 1 = 2\text{V}$

2. **(1)**

W_1 = 1000N ⇒ Weight of person at surface of earth

W_2 = 400N ⇒ Weight of person at surface of Mars.

There will be a neutral point-N where weight will be zero because at this point net gravitational field due to earth and mars is zero.



3. **(3)**

Total flux through the cube = $(q/\varepsilon^0) \times 1/8 = q/8\varepsilon^0$

Total flux through one "outer" face of the cube

$= (q/8\varepsilon^0) \times 1/3 = q/24\varepsilon^0$

[Because there is flux only through 3 faces]

Hence, total flux through shaded area

$\phi_T = (q/24\varepsilon_0 + q/24\varepsilon_0) \times 1/2$

[half of each face is shaded]

$\phi_T = q/24\varepsilon 0$

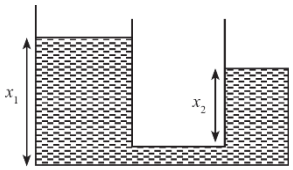
4. **(3)**

Energy in the initial stage

$$U_i = Sx_1 dg\frac{x_1}{2} + Sx_2 dg\frac{x_2}{2}$$

Energy in the final stage

$$U_f = \frac{S(x_1 + x_2)^2 gd}{4}$$



Change in energy during the process,

$$U_i - U_f = \frac{gdS}{4}\left\{2x_1^2 + 2x_2^2 - (x_1 + x_2)^2\right\}$$

$$= \frac{gdS}{4}(x_1 - x_2)^2$$

5. **(1)**

Potential of a conducting sphere is given by,

$$V = \frac{KQ}{R}$$

Given $V_1 = V_2$ and $R_1 = R_2$

$\therefore Q_1 = Q_2$

6. **(3)**

$$\because T = 2\pi\sqrt{\frac{l}{g}}$$

$$\Rightarrow T^2 \alpha l$$

7. **(3)**

Stress = $Y \times$ strain

$$\Rightarrow \frac{T_1}{A} = Y_1 \times \frac{(l_1 - l)}{l} \Rightarrow Y_1 = \frac{T_1 l}{A(l_1 - l)}$$

$$\frac{T_2}{A} = Y_2 \times \frac{(l_2 - l)}{l} \Rightarrow Y_2 = \frac{T_2 l}{A(l_2 - l)}$$

$$\because y_1 = y_2$$

$$\frac{T_1 l}{A(l_1 - l)} = \frac{T_2 l}{A(l_2 - l)}$$

$$\frac{T_1}{T_2} = \frac{l_1 - l}{l_2 - l}$$

$$T_1 l_2 - T_1 l = T_2 l_1 - T_2 l$$

$$l = \frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$$

8. **(3)**

Given, number of mole, n = 1

$$W_{AB} = nRT\ln\frac{V_2}{V_1} = 1 \times RT\ln\frac{2V_1}{V_1} = RT\ln 2$$

$W_{BC} = 0$

$$W_{CA} = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{\frac{P_1}{4} \times 2V_1 - P_1 \times V_1}{\gamma - 1}$$

$$\Rightarrow W_{CA} = \frac{\frac{P_1V_1}{2} - P_1V_1}{\gamma - 1} = \frac{-P_1V_1}{2(\gamma - 1)} = \frac{-1RT}{2(\gamma - 1)}$$

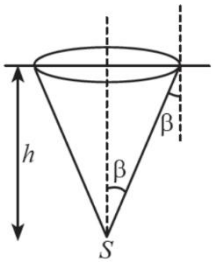
$$\Rightarrow W_{CA} = \frac{-RT}{2(\gamma - 1)}$$

$$\therefore W_{\text{total}} = W_{AB} + W_{BC} + W_{CA} = RT\left[\ln 2 - \frac{1}{2(\gamma - 1)}\right]$$

9. **(2)**

Suppose the cone of light formed on the surface of water, as shown in the figure, β is the crtical angle

$$\sin\beta = \frac{3}{4},\ \cos\beta = \frac{\sqrt{7}}{4}$$

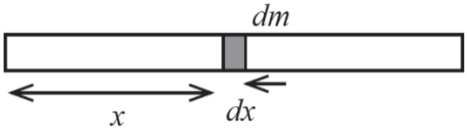


We know, Solid angle $d\Omega = 2\pi R^2 (1-\cos\beta)$

Percentage of limit $= \dfrac{2\pi R^2(1-\cos\beta)}{4\pi R^2}\times 100$

$= \dfrac{1-\cos\beta}{2}\times 100 = \left(\dfrac{4-\sqrt{7}}{8}\right)\times 100 \simeq 17\%$

10. (2)



$dm = \lambda dx = \lambda_0\left(1+\dfrac{x}{L}\right)dx$

$M = \lambda_0\int_0^L\left(1+\dfrac{x}{L}\right)dx = \lambda_0\ L + \lambda_0\dfrac{L}{2} = \dfrac{3\lambda_0 L}{2}$ .. (i)

$dI = dmx^2 = \lambda_0\left(1+\dfrac{x}{L}\right)dx\times x^2$

The moment of inertia about an axis passing through A and perpendicular to the road,

$I = \lambda_0\left\{\int_0^L x^2dx + \dfrac{1}{L}\int_0^L x^3dx\right\} = \lambda_0\left\{\dfrac{L^3}{3}+\dfrac{L^3}{4}\right\}$

$I = \dfrac{7\lambda_0 L^3}{12} = \dfrac{7}{12}\left(\dfrac{2M}{3L}\right)L^3 = \dfrac{7}{18}ML^2$

11. (2)

$B_A = \dfrac{(\mu_0)(2)}{(2)(2)}\dfrac{3(\pi/2)}{2\pi}$ …(i)

$B_B = \dfrac{(\mu_0)(3)}{(2)(4)}\dfrac{5\pi/3}{2\pi}$ …. (ii)

From (i) and (ii)

$\dfrac{B_A}{B_B} = \dfrac{6}{5}$

12. (1)

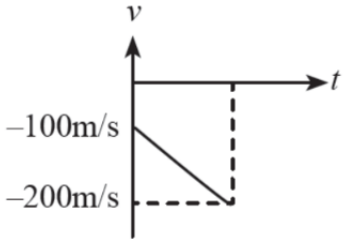
From Kinematics equations.

$v = u + at$

$v = -100 - 10t$

at, $t = 0$, $v = -100 - 10\times 0 = -100$ m/s

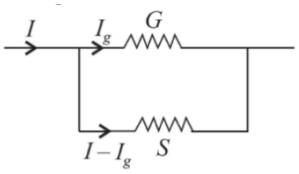
at $t = 10$s, $v = -100 - 10\times 10 = -200$ m/s



13. (2)

In galvanometer

$I_gG = (I - I_g)\times S$



$\dfrac{I_g}{I} = \dfrac{S}{S+G} \Rightarrow \dfrac{1}{3} = \dfrac{S}{S+G} \Rightarrow G = 2S$

14. (1)

$\dfrac{dx}{dt}$ = slope ≥ 0 always increasing

A. $\dfrac{dx}{dt} < 0$; and at $t\to\infty \dfrac{dx}{dt}\to 0$

B. $\dfrac{dx}{dt} > 0$ for first half $\dfrac{dx}{dt} < 0$ for second half.

C. $\dfrac{dx}{dt}$ = constant

15. (3)

$\vec{P}_x = \dfrac{h}{\lambda_x}\hat{i}$ & $\vec{P}_y = \dfrac{h}{\lambda_y}\hat{j}$

Using momentum conservation

$\vec{P} = \vec{P}_x + \vec{P}_y$

$\dfrac{h}{\lambda} = \dfrac{h}{\lambda_x} - \dfrac{h}{\lambda_y}$, $\dfrac{1}{\lambda} = \dfrac{1}{\lambda_x} - \dfrac{1}{\lambda_y}$

$|\lambda| = \left|\dfrac{\lambda_x\lambda_y}{\lambda_x - \lambda_y}\right|$

16. (2)

$E \propto \dfrac{1}{r}$, $r \propto \dfrac{1}{m}$

$E \propto m$ b

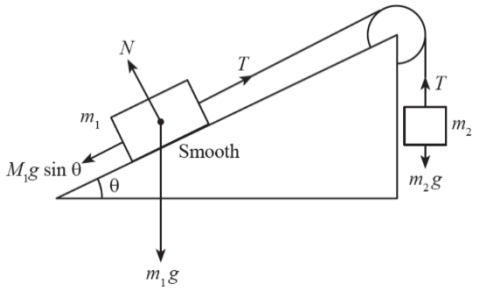
Ionization potential $= 13.6\times\dfrac{(m_\mu)\text{eV}}{(m_e)}$

= 13.6 × 207 eV = 2815.2 eV

17. (2)

As m_1 is in rest

T = m_2g = 30 N



$T = m_1g\sin\theta \Rightarrow 50\sin\theta = 30$

$\Rightarrow \sin\theta = \dfrac{3}{5} \Rightarrow \theta = 37°$

$\therefore N = 50\cos\theta = 50\cos 37° = 50\times\dfrac{4}{5} = 40N$

18. (4)

Fringe width, $\beta = \frac{D\lambda}{d}$

$\lambda_1 = 5000$ Å

$\beta_1 = \frac{D}{d}(5000\times10^{-10}) = 5\times10^{-4}$ m .... (i)

$\beta_2 = \frac{D}{(2d)}(6000\times10^{-10}) = x(\text{let})$ ... (ii)

Divide (ii) and (i)

$\frac{\beta_2}{\beta_1} = \frac{3000\times10^{-10}}{5000\times10^{-10}} = \frac{x}{5\times10^{-4}}$

$x = 3 \times 10^{-4}$ m or 0.3 mm

19. (4)

By law of conservation of energy

$\frac{1}{2}kx^2 = (m_1s_1 + m_2s_2)\Delta T$

$\Delta T = \frac{16\times10^{-2}}{4384} = 3.65\times10^{-5}K$

20. (2)

Δm should be positive

$(m_p + m_n) > m_d$

$\Rightarrow$ only 2 is possible

21. (105)

It is given that 315 Hz and 420 Hz are two resonant frequencies, let these be nth and $(n + 1)^{th}$ harmonies, then we have $\frac{nv}{2l} = 315$ and

$(n+1)\frac{v}{2l} = 420$

$\Rightarrow \frac{n+1}{n} = \frac{420}{315}$

$\Rightarrow n = 3$

Hence $3\times\frac{v}{2l} = 315 \Rightarrow \frac{v}{2l} = 105\,\text{Hz}$

The lowest resonant frequency is when $n = 1$

Therefore lowest resonant frequency = 105 Hz.

22. (4)

The inner sphere is grounded, hence its potential is zero. The net charge on isolated outer sphere is zero.

Let the charge on inner sphere be q'.

$\therefore$ Potential at centre of inner sphere is

$= \frac{1}{4\pi\varepsilon_0}\frac{q'}{a} + 0 + \frac{1}{4\pi\varepsilon_0}\frac{q}{4a} = 0$

$\therefore q' = -\frac{q}{4}$

23. (5)

From Kirchoff's current law,

$i_3 = i_1 + i_2 = 3\sin\omega t + 4\sin(\omega t + 90°)$

$\Rightarrow i_3 = i_0\sin(\omega t + \phi)$

Where $i_0 = \sqrt{3^2 + 4^2 + 2(3)(4)\cos90°}$

and $\tan\theta = \frac{4\sin90°}{3 + 4\cos90°} = \frac{4}{3}$

$\therefore i_3 = 5\sin(\omega t + 53°)$

24. (5)

Energy of photon (E) is given by $E = \frac{hc}{\lambda}$

Number of photons of wavelength λ emitted in t second from laser of power P is given by

$n = \frac{Pt\lambda}{hc}$

$\Rightarrow n = \frac{2\times\lambda}{hc} = \frac{2\times10^{-3}\times5\times10^{-7}}{2\times10^{-25}} (\because t = 1S)$

$\Rightarrow n = 5 \times 10^{15}$

25. (2)

From stefan's law, the energy radiated per second is given by $E = e\sigma T^4A$

Here, T = temperature of the body

A = surface area of the body

For same material e is same. σ is stefan's constant

Let T_1 and T_2 be the temperature of two spheres.

A_1 and A_2 be the area of two spheres

$\therefore \frac{E_1}{E_2} = \frac{T_1^4A_1}{T_2^4A_2} = \frac{T_1^4 4\pi r_1^2}{T_2^4 4\pi r_2^2}$

$= \frac{(4000)^4\times1^2}{(2000)^4\times4^2} = \frac{1}{1}.$

SECTION-II (CHEMISTRY)

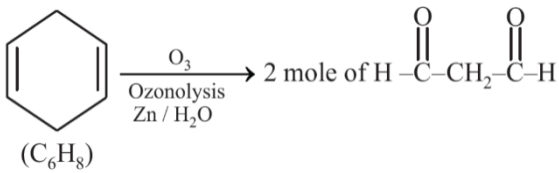
26. (1)

Carbylamine reaction is given by primary amines only.

27. (4)

Hydrocarbon 'X ' is cyclohexa –1,4–diene

The reaction is given below:



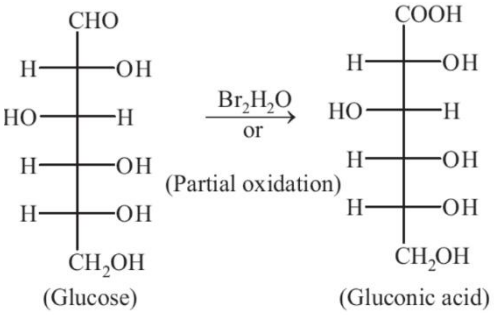
(C_6H_8)

(Cyclohexa -1, 4-diene)

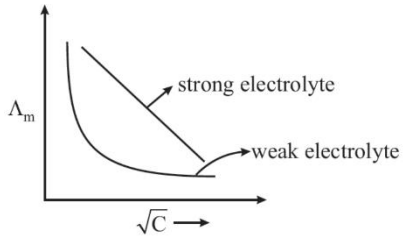
28. (3)

Water will be neutral as $[H^+] = [OH^-]$

29. **(3)**



30. **(3)**

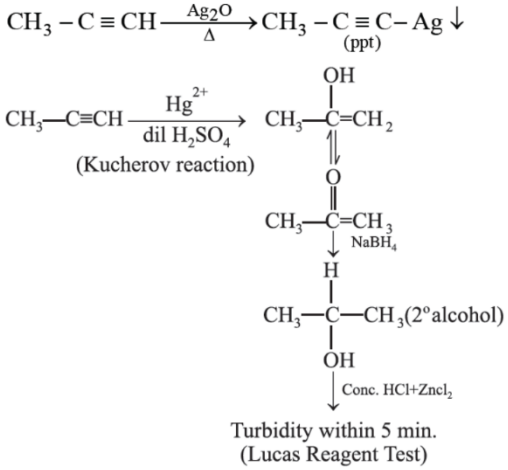


CH_3COOH is weak electrolyte which is given in only option (c).

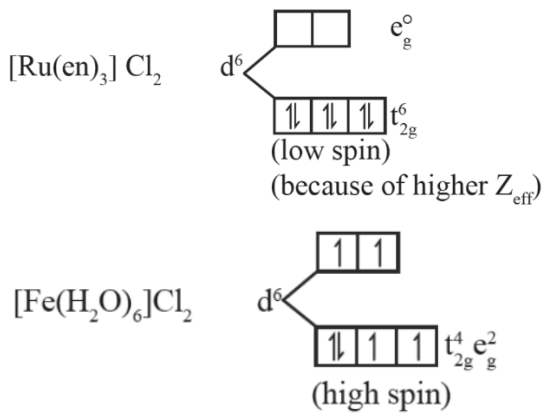
31. **(2)**

pOH α Acidic strength

32. **(2)**

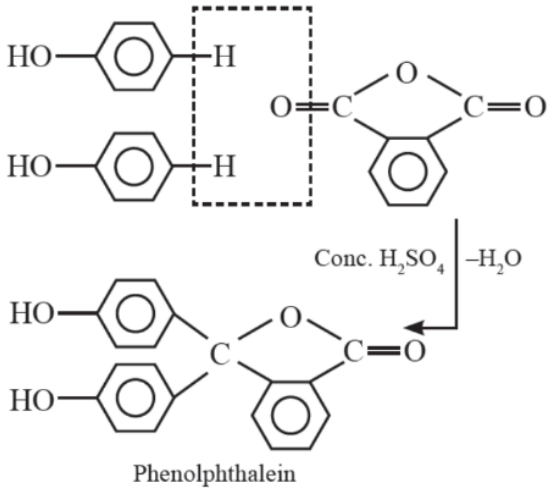


33. **(1)**



34. **(1)**

For the given conditions, the possible reaction is represented as,



Hence, the compound X and Y respectively are carbolic acid and phenolphthalein.

35. **(4)**

- Metallic character decreases from left to right and non metallic character increases as one move from left to right.
- Both I. E. and electron gain enthalpy increases as one move from left to right.

36. **(4)**

$$CH_4 + I_2 \rightleftharpoons CH_3I + HI$$

It is reversible reaction due to reducing nature of HI but it can be converted to irreversible by adding oxidizing agents like, HNO_3, HIO_3, HgO, etc which can oxidize HI into I_2.

37. **(2)**

$\Delta T_b = K_b m$

$\Delta T_b = K_b \times 1 \Rightarrow 2 = K_b$

$\Delta T_f = K_f m$

$2 = 2K_f = K_f$

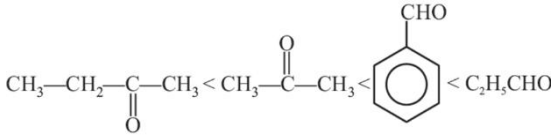
$$\frac{K_f}{K_b} = \frac{1}{2}$$

38. **(1)**

The given electron configuration is of Ga and in 5th period, diagonally situated element will be Sn with respect to Ga. Hence, electronic configuration of Sn is: $[Kr]\ 4d^{10}\ 5s^2\ 5p^2$

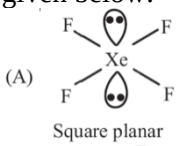
39. **(2)**

Increasing hindrence and electron releasing group (ERG) attached on carboxyl group decrease the reactivity. Correct order of nucleophilic addition reaction is:

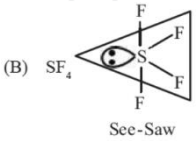


40. (4)

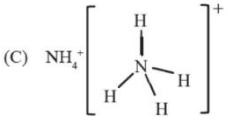
According to the VSEPR theory, the shape of the given compounds along with their shapes name is given below:



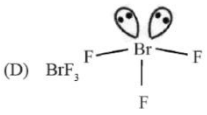
(A) Square planar



(B) See-Saw

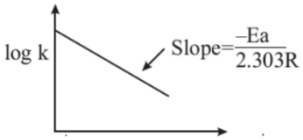


(C) Tetrahedral



(D) Bent T-Shaped

41. (2)



$k = A.e^{-Ea/RT}$

$$\underset{y}{\log k} = \underset{c}{\log A} - \underbrace{\frac{Ea}{2.303R}}_{m}.\underset{x}{\frac{1}{T}} \quad \ldots (i)$$

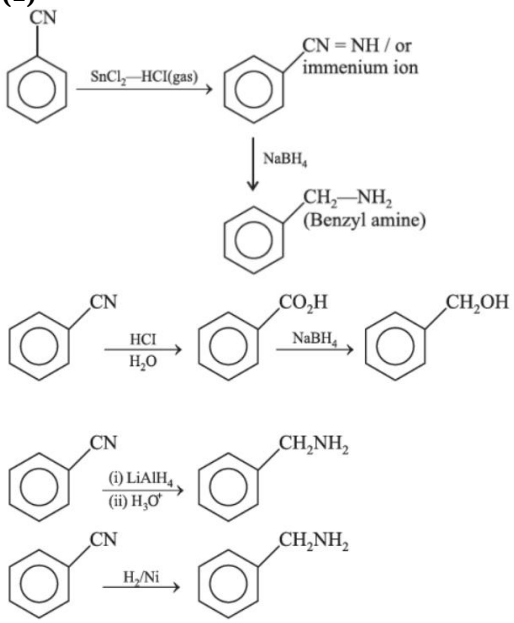
Higher is Ea, stronger is the temperature dependence of k (i.e. steeper the slope)

Put $E_a = 0$

in eq. (i)

log K = log A

42. (1)



43. (2)

The decreasing order of priority of functional group is,

$-SO_3H > -COCl > -CONH_2 > -CN$

44. (3)

$\Delta G^0 = -RT \ln K$ and $\Delta G^0 = \Delta H^0 - T\Delta S^0$

45. (1)

$CrO_3 \rightarrow +6$; $Fe_2O_3 \rightarrow +3$; $MnO_2 \rightarrow +4$; $V_2O_5 \rightarrow +5$; $Cu_2O \rightarrow +1$

Hence, correct order is: E < B < C < D < A

46. (7)

$2KMnO_4 + KI + H_2O \rightarrow 2KOH + 2MnO_2 + KIO_3$

3.5 moles of potassium iodide requires 7 mol of potassium permanganate.

47. (4)

The reaction takes place as follows:

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

$$\text{Moles of Zn} = \frac{11.5}{65.4}\text{moles}$$

According to the reaction,

1 mole of Zn produces 1 mole of H_2 gas

Thus, $\left(\frac{11.5}{65.4}\right)$ moles of Zn produce $\left(\frac{11.5}{65.4}\right)$ moles of H_2 gas.

At STP

1 mol of H_2 produces 22.4 L of volume.

Hence, for $\left(\frac{11.5}{65.4}\right)$ moles of H_2, Volume

$$= 22.4 \times \left(\frac{11.5}{65.4}\right) = 3.94\,\text{L} = 4\text{L}$$

48. (9)

$K_b = 10^{-5}$, $pK_b = 5$

$BOH + HCl \rightarrow BCl + H_2O$

At half neutralization, 50% of the base is converted to its salt, with strong acid HCl, it forms a basic buffer.

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

$$pOH = 5 + \log\frac{[salt]}{[base]}$$

[salt] = [base]

pOH = 5

pH = 14 – pOH = 9

49. (5)

$$4H^+ + MnO_4^- \rightleftharpoons MnO_2 + 2H_2O$$

$$\Delta G^\circ = -3F\left(\frac{1.5\times 5 - 2\times 1.25}{3}\right) = -5F$$

x = 5

50. (270)

For the given reaction, enthalpy change is,

$\Delta H = \Delta U + \Delta n_g RT$

$= 2.1 \times 10^3 + 2 \times 2 \times 300 = 3300$ cal.

$\Delta G = \Delta H - T\Delta S$

$= 3300 - 300 \times 20 = -2700$ cal.

$= -2.7$ kcal

SECTION-III (MATHEMATICS)

51. (1)

Given α and β are the roots of the equation $x^2 - x - 1 = 0$

$\Rightarrow \alpha + \beta = 1$ and $\alpha\beta = -1$

$\alpha^2 = \alpha + 1$, $\beta^2 = \beta + 1$

$\alpha^5 = 5\alpha + 3$

$\beta^2 = 5\beta + 3$

_____.

$p_5 = 5(\alpha + \beta) + 6 = 5(1) + 6$

$p_5 = 11$ and $p_2 = \alpha 2 + \beta 2 = \alpha + 1 + \beta + 1$

$p_2 = 3$ and $p_3 = \alpha 3 + \beta 3 = 2\alpha + 1 + 2\beta + 1$

$= 2(1) + 2 = 4$

$p_2 \times p_3 = 12$ and $p_5 = 11 \Rightarrow p_5 \neq p_2 \times p_3$

52. (2)

$$A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$$

Given: $A^3 = A$

Now, $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$

$$A^2 = \begin{bmatrix} a+2 & 2c & 3 \\ 3 & a+3c & 2a \\ ac & 1 & 2+3c \end{bmatrix}$$

and $A^3 = \begin{bmatrix} a+2 & 2c & 3 \\ 3 & a+3c & 2a \\ ac & 1 & 2+3c \end{bmatrix}\begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ 1 & c & 0 \end{bmatrix}$

$$A^3 = \begin{bmatrix} 2ac+3 & a+2+3c & 2a+4+6c \\ a(a+3c)+2a & 3+2ac & 6+3a+9c \\ a+2+3c & ac+c(2+3c) & 2ac+3 \end{bmatrix}$$

Given $A^3 = A$

$2ac + 3 = 0$ … (i)

$a + 2 + 3c = 1$

$\Rightarrow a + 1 + 3c = 0$

$a + 1 - \frac{9}{2a} = 0 \quad \left[\because n = \frac{-3}{2a} \text{ from eqation(i)}\right]$

$2a^2 + 2a - 9 = 0$

$f(1) < 0, f(2) > 0 \Rightarrow a \in (1, 2]$

$n = 2$

53. (1)

$$x^2\left(\sqrt{x} + \frac{\lambda}{x^2}\right)^{10}$$

Consider constant term

$${}^{10}C_r \left(\sqrt{x}\right)^{10-r}\left(\frac{\lambda}{x^2}\right)^r$$

$\Rightarrow \frac{10 - r}{2} - 2r = 0$

$\Rightarrow 10 - 5r = 0$

$\Rightarrow r = 2$

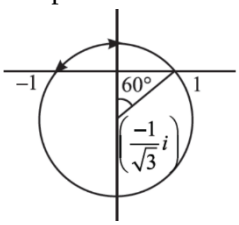
${}^{10}C_2 \times \lambda^2 = 720 \Rightarrow \lambda = 4$

54. (2)

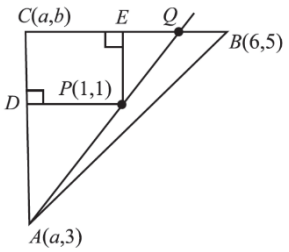
$$\left|\frac{z+1}{z-1}\right| < 1 \Rightarrow |z+1| < |z-1| \Rightarrow \text{Re}(z) < 0 \text{ and}$$

$$\arg\left(\frac{z-1}{z+1}\right) = \frac{2\pi}{3}$$

is a part of circle as shown.



55. (2)



Since, point *D* is mid-point of AC.

$$\therefore \text{point } D\left(a, \frac{b+3}{2}\right)$$

Line *AC* is a vertical line. Therefore, line $\perp$ to it will be parallel to x-axis

$\therefore$ Slope of $DP = 0$

$$\Rightarrow \frac{1 - \frac{b+3}{2}}{1-a} = 0$$

$\Rightarrow b = -1$

Also, point *E* is mid-point of *BC*.

$$\therefore Pt\ E\left(\frac{a+b}{2}, \frac{b+5}{2}\right) \text{ where } b = -1$$

Line *EP* is $\perp$ to *BC*.

$\therefore$ Slope of *EP* × Slope of *BC* = – 1

$$\Rightarrow \frac{1 - \frac{b+5}{2}}{1 - \frac{a+b}{2}} \times \frac{5-b}{b-a} = -1$$

$\Rightarrow 12 = (a + 1)(a - 3)$

$\Rightarrow a = 5$ or -3

Since, $ab > 0$ (given) $\Rightarrow a = -3$

Line AP : $x + 2y = 3$ … (i)

Line *BC*:

$B(-1, 5)$, $C(-3, -1)$

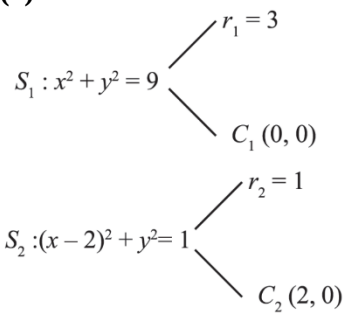
$3x - y = -8$ …. (ii)

On solving eq. (i) and (ii), we get

$\Rightarrow x = \frac{-13}{7}$ and $y = \frac{17}{7}$

$\therefore k_1 + k_2 = x + y = \frac{4}{7}$

56. (4)



$\therefore C_1C_2 = r_1 - r_2$ [Given, circle are touching internally]

Let centre of variable circle is at $C(h, k)$ & radius = r

$CC_2 = 3 - r$

$CC_2 = 1 + r$

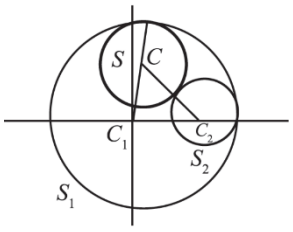
$CC_1 + CC_2 = 4$ (constant)

$\therefore$ Locus of C is an ellipse.

$\sqrt{(h-2)^2 + k^2} = 4 - \sqrt{h^2 + k^2}$

$\Rightarrow 3h^2 + 4k^2 - 6h - 9 = 0$

Point $\left(2, \pm\frac{3}{2}\right)$ satisfying the above expression



57. (1)

Given, $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}\,dx = g(x) + c$

Put $x = \cos 2\theta \Rightarrow dX = -2\sin 2\theta.\, D\theta$

$= \int \frac{1}{\cos 2\theta}\tan\theta(-4\sin\theta.\cos\theta)d\theta$

$= \int \frac{1}{\cos 2\theta}(-4\sin^2\theta)\,d\theta$

$= -2\int \frac{1-\cos 2\theta}{\cos 2\theta}d\theta$

$= -\frac{2}{2}\ln\left|\frac{1-\sin 2\theta}{\cos 2\theta}\right| + \cos^{-1}x + c$

$\underbrace{= \ln\left|\frac{1-\sqrt{1-x^2}}{x}\right| + \cos^{-1}x + c}_{g(x)}$

$\because g(1) = 0 \Rightarrow c = 0$

$g(x) = \ln\left|\frac{1-\sqrt{1-x^2}}{x}\right| + \cos^{-1}x$

$g\left(\frac{1}{2}\right) = \ln\left|2-\sqrt{3}\right| + \frac{\pi}{3} \Rightarrow g\left(\frac{1}{2}\right) = \ln\left|\frac{\sqrt{3}-1}{\sqrt{3}+1}\right| + \frac{\pi}{3}$

58. (4)

$\sum_{k=1}^{20}(1+2+3+....+k) = \sum_{k=1}^{20}\frac{k(k+1)}{2}$

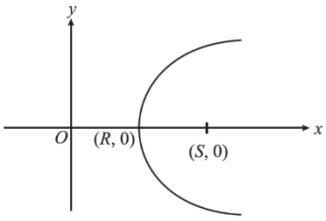
$= \frac{1}{2}\sum_{k=1}^{20}(k^2+k) = \frac{1}{2}\left[\frac{20(21)(41)}{6} + \frac{(20)(21)}{2}\right]$

$= \frac{1}{2}\left[\frac{420\times 41}{6} + \frac{20\times 21}{2}\right]$

$= \frac{1}{2}[2870 + 210] = 1540$

59. (4)

Length of latus rectum =4 (distance between vertex and focus) = 4 (S - R)



60. (2)

$f(g(x)) = \begin{bmatrix} x^3+2 & ; & x^3 < 0 \,\&\, x < 1 \\ x^6 & ; & x^3 \geq 0 \,\&\, x < 1 \\ 3x & ; & 3x-2 < 0 \,\&\, x \geq 1 \\ (3x-2)^2 & ; & 3x-2 \geq 0 \,\&\, x \geq 1 \end{bmatrix}$

$\Rightarrow fog(x) = \begin{bmatrix} x^3+2 & ; & x < 0 \\ x^6 & ; & 0 \leq x < 1 \\ (3x-2)^2 & ; & x \geq 1 \end{bmatrix}$

$L.H.L = \lim_{h\to 0^-} fog(x) = 2$

$R.H.L = \lim_{h\to 0^+} fog(x) = 0$

$\therefore fog(x)$ is discontinuous at $x = 0$

Hence, $fog(x)$ is not differentiable at $x = 0$

Now, $L.H.D = \lim_{h\to 0^-}\frac{f(1+h) - f(1)}{h}$

$= \lim_{h\to 0^-}\frac{(1+h)^6 - 1^6}{h}$

$= \lim_{h\to 0^-}\frac{(1+h)^6 - 1^6}{h}$

$= \lim_{h\to 0^-}\frac{(1+h)^6 - 1^6}{h}$

$$= \lim_{h\to 0^-} \frac{6(1+h)^5}{i} = 6$$

$$= \lim_{h\to 0^+} \frac{f(i+h)^5 - f(i)}{h}$$

$$= \lim_{h\to 0^+} \frac{(3(1+h)-2)^2 - (3-2)^2}{h} = 6$$

Thus L.H.D = R.H.D

Hence, *fog* (x) is differentiable at $x = 1$

61. (2)

Four digit numbers greater than 7000

$= 2 \times 4 \times 3 \times 2 = 48$

Five digit number = 5! = 120

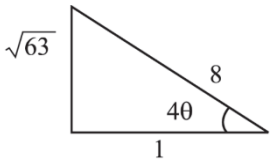
Total number greater than 7000

= 120 + 48 = 168

62. (1)

Given, $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$

Put $\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8} = \theta$



$$\Rightarrow \sin 4\theta = \frac{\sqrt{63}}{8}$$

$$\therefore \cos 4\theta = \frac{1}{8}$$

$$2\cos^2 2\theta = \frac{1}{8} + 1 \Rightarrow \cos 2\theta = \frac{3}{4}$$

$$\Rightarrow \frac{1-\tan^2\theta}{1+\tan^2\theta} = \frac{3}{4}$$

(adding 1 both side)

$$\Rightarrow \frac{2}{\sec^2\theta} = \frac{7}{4}$$

$$\Rightarrow \cos^2\theta = \frac{7}{8} \Rightarrow \sec^2\theta = \frac{8}{7}$$

$$\Rightarrow \tan^2\theta = \frac{8}{7} - 1 = \frac{1}{7}$$

$$\Rightarrow \tan\theta = \frac{1}{\sqrt{7}}$$

63. (4)

$-6 < n^2 - 10n + 19 < 6$

$\Rightarrow n^2 - 10n + 25 > 0$ and $n^2 - 10n + 13 < 0$

$\Rightarrow (n-5)^2 > 0$ and $n \in \left[5 - 2\sqrt{3}, 5 + 2\sqrt{3}\right]$

$n \in R - \{5\}$ …. (i)

$n = \{2, 3, 4, 5, 6, 7, 8\}$ … (ii)

Taking intersection of (i) & (ii), we get

$\Rightarrow n = 2, 3, 4, 6, 7, 8$

$\therefore$ Number of values of $n = 6$

64. (3)

Given, $E : \frac{x^2}{4} + \frac{y^2}{2} = 1$

Let a point on it is $P\left(2\cos\theta, \sqrt{2}\sin\theta\right)$

$\therefore M(h, k)$ be mid point of P & $A(4, 3)$

Then $h = \frac{2\cos\theta + 4}{2} \Rightarrow \cos\theta = (h-2)$

$$\& k = \frac{\sqrt{2}\sin\theta + 3}{2} \Rightarrow \sin\theta = \frac{2k-3}{\sqrt{2}}$$

$$\Rightarrow (h-2)^2 + \left(\frac{2k-3}{\sqrt{2}}\right)^2 = 1$$

Required locus $(x-2)^2 + \frac{\left(y - \frac{3}{2}\right)^2}{\frac{1}{2}} = 1$

$$\therefore e = \sqrt{1 - \frac{1}{2}} = \frac{1}{\sqrt{2}}$$

65. (1)

$\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and

$\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$

Now, $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$

$\Rightarrow (\vec{d} - \vec{c}) \times \vec{b} = 0$

$\Rightarrow (\vec{d} - \vec{c})$ is parallel to $\vec{b}$

$\Rightarrow \vec{d} = \vec{c} + k\vec{b}$

Also Also $\vec{d}.\vec{a} = 24$

$\Rightarrow (\vec{c} + k\vec{b}).\vec{a} = 24$

$$k = \frac{24 - \vec{a}.\vec{c}}{\vec{b}.\vec{a}} = \frac{24-6}{9} = 2$$

So, $\vec{d} = \vec{c} + 2(\vec{b})$

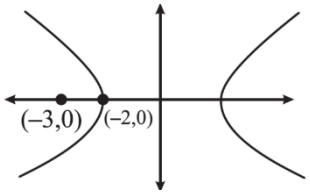
$= 8\hat{i} - 5\hat{j} + 18\hat{k}$

$\Rightarrow |\vec{d}|^2 = 64 + 25 + 324 = 413$

66. (4)

Equation of hyperbola is $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$ and $ae = 3$

We know that $a^2e^2 = a^2 + b^2$



$9 = 4 + b^2 \Rightarrow b^2 = 5$

Hence, equation of hyperbola is $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$

Hence, $\left(6, 5\sqrt{2}\right)$ does not lie on $\frac{x^2}{4} - \frac{y^2}{5} = 1$

67. (1)

$$\left[\frac{x}{\sqrt{x^2 - y^2}} + e^{y/x}\right] x\frac{dy}{dx}$$

$$= x + \left[\frac{x}{\sqrt{x^2 - y^2}} + \sqrt{e}^{y/x}\right] y$$

$$\Rightarrow e^{y/x}[xdy - ydx] = xdx + \frac{x}{\sqrt{x^2 - y^2}}(ydx - xdy)$$

$$\Rightarrow e^{y/x}\left[\frac{xdy - ydx}{x^2}\right] = \frac{dx}{x} - \frac{x}{\sqrt{x^2 - y^2}}\frac{(xdy - ydx)}{x^2}$$

$$\Rightarrow e^{y/x} d(y - x) = \frac{dx}{x} - \frac{d(y - x)}{\sqrt{1 - (y - x)^2}}$$

Integrating both sides, we get

$$e^{y/x} = \ln(x) - \sin^{-1}\left(\frac{y}{x}\right) + c$$

It passes through (1, 0),
So, $e^{\circ} = 0 - \sin^{-1}(0) + c$
$\Rightarrow 1 = c$

Hence, $e^{y/x} = \ln(x) - \sin^{-1}\left(\frac{y}{x}\right) + 1$

It also passes through $(2\alpha, \alpha)$.
$\therefore e^{1/2} = \ln 2\alpha - \sin^{-1}(1/2) + 1.$

$$\ln 2\alpha = \sqrt{e} - 1 + \frac{\pi}{6}$$

$$\Rightarrow \alpha = \frac{1}{2} e^{\left(\frac{\pi}{6} + \sqrt{e} - 1\right)}$$

$$\Rightarrow \alpha = \frac{1}{2}\exp\left(\sqrt{e} - 1 + \frac{\pi}{6}\right)$$

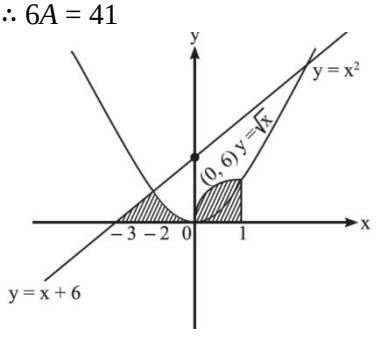
68. (3)

Area of shaded region,

$$A = \int_{-3}^{-2} (x + 6)dx + \int_{-2}^{0} x^2 dx + \int_{0}^{1} \sqrt{x}dx$$

$$= \frac{7}{2} + \frac{8}{3} + \frac{2}{3} = \frac{41}{6}$$

$\therefore 6A = 41$



69. (2)

We have, $f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$

$$\Rightarrow f'(x) = 2x\left(\frac{x^4 - 5x^2 + 4}{2 + e^{x^2}}\right) = 0$$

$\Rightarrow x = 0$ or $(x^2 - 4)(x^2 - 1) = 0$
$\Rightarrow x = 0$ or $x = \pm 2, \pm 1$
Thus, $f'(x)$ can be written as,

$$f'(x) = \frac{2x(x+1)(x-1)(x+2)(x-2)}{e^{x^2+2}}$$

Since, $f'(x)$ changes sign from positive to negative at $x = -1, 1$.
So, number of local maximum points = 2
$\because f'(x)$ changes sign from negative to positive at x
$= -2, 0, 2$ So, number of local minimum points = 3
$\therefore m = 2, n = 3$

70. (4)

Given trigonometric equation is
$\sin 12° + \sin 12° - \sin 72°$
$= \sin 12° - 2\cos 42° \sin 30°$

$$\left[\sin C - \sin D = 2\cos\left(\frac{C + D}{2}\right)\sin\left(\frac{C - D}{2}\right)\right]$$

$= \sin 12° - \sin(90° - 42°) = \sin 12° - \sin 48°$
$= -2\cos 30° \sin 18°$

$$= -2 \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{5} - 1}{4} = \frac{\sqrt{3}}{4}\left(1 - \sqrt{5}\right)$$

71. (398)

Given, $7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, \ldots$
$t_r = (3r + 4)(2r + 6) = 6r^2 + 26r + 24$

$$\therefore \sum_{r=1}^{10} t_r = \frac{6 \times 10 \times 11 \times 21}{6} + \frac{26 \times 10 \times 11}{2}$$

$+ 24 \times 10 = 10 \times 398$

$$\therefore \text{Mean} = \frac{10 \times 398}{10} = 398$$

72. (18)

If the lines $\frac{x - 1}{2} = \frac{2 - y}{-3} = \frac{z - 3}{\alpha}$

And $\frac{x - 4}{5} = \frac{y - 1}{2} = \frac{z}{\beta}$ intersect

Point on first line (1, 2, 3) and point on second line (4, 1, 0).

Vector joining both points is $-3\hat{i} + \hat{j} + 3\hat{k}$

Now vector along first line is $2\hat{i} + 3\hat{j} + \alpha\hat{k}$

Also vector along second line is $5\hat{i}+2\hat{j}+\beta\hat{k}$

Now these three vectors must be coplanar

$$\Rightarrow \begin{vmatrix} 2 & 3 & \alpha \\ 5 & 2 & \beta \\ -3 & 1 & 3 \end{vmatrix} = 0$$

$\Rightarrow 2(6-\beta)-3(15+3\beta)+\alpha(11)=0$

$\Rightarrow \alpha-\beta=3$

Now, $\alpha=3+\beta$

Given expression is $8\alpha\beta = 8(3+\beta).\beta = 8(\beta^2+3\beta)$

$$=8\left(\beta^2+3\beta+\frac{9}{4}-\frac{9}{4}\right)=8\left(\beta+\frac{3}{2}\right)^2-18$$

So magnitude of minimum value of $8\alpha\beta = 18$

73. (4)

$$f(x)\int \frac{\frac{5}{x^6}+\frac{7}{x^8}}{\left(2+\frac{1}{x^5}+\frac{1}{x^7}\right)^2}dx$$

$$=\frac{1}{2+\frac{1}{x^5}+\frac{1}{x^7}}+c$$

$$\Rightarrow f(x)=\frac{x^7}{2x^7+x^2+1}\ (\because f(0)=0)$$

$$\Rightarrow f(1)=\frac{1}{4}$$

74. (5)

Given, $P(E)<\frac{1}{2}$

General form : $\sum_{r=n}^{8} {}^8C_r\left(\frac{1}{2}\right)^{8-r}\left(\frac{1}{2}\right)^r<\frac{1}{2}$

$$\Rightarrow \sum_{r=n}^{8} {}^8C_r\left(\frac{1}{2}\right)^8<\frac{1}{2}$$

$$\Rightarrow {}^8C_n+{}^8C_{n+1}+.....{}^8C_8<2^7$$

$$\Rightarrow 256-({}^8C_0+{}^8C_1+.....{}^8C_{n-1})<128$$

$$\Rightarrow {}^8C_0+{}^8C_1+{}^8C_{n-1}>128$$

$\Rightarrow n-1\geq 4$

Hence, $n\geq 5$

75. (1)

Coefficient of x^9 in $\left(\alpha x^3+\frac{1}{\beta x}\right)^{11} = {}^{11}C_6.\frac{\alpha^5}{\beta^6}$

Coefficient of x^{-9} in $\left(\alpha x-\frac{1}{\beta x^3}\right)^{11} = -{}^{11}C_5.\frac{\alpha^6}{\beta^5}$

$\because$ Both coefficient are equal

$$\therefore {}^{11}C_6.\frac{\alpha^5}{\beta^6}=-{}^{11}C_5.\frac{\alpha^6}{\beta^5}\Rightarrow\frac{1}{\beta}=-\alpha$$

$\Rightarrow \alpha\beta=-1$

$\Rightarrow (\alpha\beta)^2=1$



PW Web/App - https://smart.link/7wwosivoicgd4

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